ILESH DHALL

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Contact

Graphic Era (Deemed to be University)
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Links: Website, Google Scholar, GitHub, LinkedIn

Research Interests

Efficient LLM inference and machine learning systems, with interests in KV-cache compression, attention mechanisms, long-context inference, model compression, and GPU-efficient deep learning.

Education

Graphic Era (Deemed to be University) — **July 2023 – Present**
B.Tech in Computer Science & Engineering, Specialization in AI and ML — CGPA: 8.85 / 10.00

Bawa Lalvani Public School — **April 2022 – March 2023**
Senior Secondary (Class XII), PCM with IT — Percentage: 82.50 / 100.00

Bawa Lalvani Public School — **April 2020 – March 2021**
Secondary (Class X) — Percentage: 95.17 / 100.00

Experience

Graphic Era (Deemed to be University), Dehradun, India — **June 2026 – July 2026**
Research Intern, Advised by Dr. Arun Chauhan
Developed GPU-efficient techniques for long-context LLM inference through KV-cache compression; integrated exact attention saliency computation into FlashAttention and evaluated memory usage, latency, context length, and model accuracy across state-of-the-art LLM benchmarks.

Conference Publications

[C1] **SkyRis: An AI-Powered IoT Framework for Real-Time Cloudburst Prediction and Alerting**.
E. Semwal, C. Bose, P. Singh, D. Sharma, D. Dhyani, S. S. Rawat, I. Dhall, R. Kumar.
International Conference on Computing, Sciences and Communications (ICCSC), 2026.

Manuscripts Under Review

[R1] **Exact Attention Saliency for Memory-Efficient Long-Context LLM Inference**.
I. Dhall, A. Chauhan.

Projects

Deep Neural Compression for Neural Style Transfer — **Oct 2024 – Dec 2024**
Performed layer-wise sensitivity analysis across 16 convolutional layers and evaluated 96 pruning configurations for VGG19-based neural style transfer. Applied sensitivity-guided magnitude pruning, achieving an 85% reduction in model size and 3.2× memory reduction, preserving stylization quality.

Metabolomics Data Analysis for Biomarker Discovery — **Apr 2025 – May 2025**
Analyzed a metabolomics dataset containing 23,000+ samples across 80+ metabolite features. Developed a preprocessing and machine learning pipeline incorporating feature pivoting, outlier capping, correlation filtering, SMOTE balancing, and ensemble classification, achieving an 89% F1-score.

AgriVerse: Multi-Agent AI Agricultural Platform — **Apr 2026 – May 2026**
Architected a multi-agent AI system with four specialized agents orchestrated through workflow automation. Built a FastAPI backend and RAG pipeline using transformer-based embeddings and ChromaDB, and deployed containerized services on GCP using a microservice architecture.

Story-Forge: Fine-Tuned LLM Story Generation Platform — **May 2026**
Fine-tuned Phi-3 Mini (3.8B) using QLoRA with 4-bit NF4 quantization and LoRA adapters. Optimized inference using Q4_K_M GGUF and llama.cpp for lightweight CPU-based deployment.

Awards & Honors

Best Paper Award, IEEE ICCSC 2026 — **2026**
Awarded for *SkyRis: An AI-Powered IoT Framework for Real-Time Cloudburst Prediction and Alerting*.

Amazon Machine Learning Summer School — **2026**
Selected among 3,000 participants from 130,000+ applicants for Amazon's competitive machine learning program.

2nd Place, alrIEEEna State-Level ML Competition — **2026**
Ranked 2nd among 100+ participants by developing a high-performing machine learning model for multi class classification.

Technical Skills

Languages: Python, SQL, C, C++
Machine Learning: PyTorch, scikit-learn, Transformers, LLMs, RAG, Fine-tuning
Model Efficiency: Quantization, Model Compression, FlashAttention, vLLM, llama.cpp
Data Analysis: NumPy, Pandas, Matplotlib, OpenCV, Feature Engineering
Systems & Cloud: FastAPI, Docker, Linux, MySQL, GCP, Git